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Control of Daughter Cell Fates during Asymmetric Division: Interaction of Numb and Notch

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Summary

During development of the Drosophila peripheral nervous system, a sensory organ precursor (SOP) cell undergoes rounds of asymmetric divisions to generate four distinct cells of a sensory organ. Numb, a membrane-associated protein, is asymmetrically segregated into one daughter cell during SOP division and acts as an inherited determinant of cell fate. Here, we show that Notch, a transmembrane receptor mediating cell–cell communication, functions as a binary switch in cell fate specification during asymmetric divisions of the SOP and its daughter cells in embryogenesis. Moreover, *numb* negatively regulates *Notch*, probably through direct protein–protein interaction that requires the phosphotyrosine-binding (PTB) domain of Numb and either the RAM23 region or the very C-terminal end of Notch. *Notch* then positively regulates a transcription factor encoded by *tramtrack* (*ttk*). This leads to Ttk expression in the daughter cell that does not inherit Numb. Thus, the inherited determinant Numb bestows a bias in the machinery for cell–cell communication to allow the specification of distinct daughter cell fates.

Introduction

In asymmetric division, an important process in the generation of cell diversity during development, a mother cell produces two daughter cells that ultimately adopt distinct fates. Two mechanisms may be responsible for making the two daughter cells different. One, the cell-intrinsic mechanism, involves an inherited determinant that is asymmetrically segregated to one daughter cell at the time of cell division. The other, the cell-extrinsic mechanism, entails communication of the daughter cells with each other or with surrounding cells (reviewed by Horvitz and Herskowitz, 1992). In the Drosophila peripheral nervous system (PNS), both mechanisms are used to generate the distinct cell types of a sensory organ (Uemura et al., 1989; Hartenstein and Posakony, 1990; Parks and Muskavitch, 1993; Rhyu et al., 1994). It thus provides an opportunity to assess how these two mechanisms interact in the process of daughter cell fate specification during asymmetric divisions.

Two main types of sensory organs in the larval and adult Drosophila PNS, the external sense (es) organ and chodotonal (cho) organ, are formed in a progressive process during embryogenesis and pupal development (Bodmer and Jan, 1987; Campos-Ortega and Hartenstein, 1985; Ghysen et al., 1986). The "proneural genes," including those in the *achaete–scute* complex and *atonal*, are first expressed in clusters of ectodermal cells to endow these cells with the competence to adopt neuronal fates (reviewed by Ghysen and Dambly-Chaudiere, 1988; Campuzano and Modolell, 1992). One sensory organ precursor (SOP) is then singled out from each cluster through a process of "mutual inhibition" mediated by "neurogenic genes" including *Notch* and *Delta* (reviewed by Campos-Ortega, 1988; Artavanis-Tsakonas and Simpson, 1991; Ghysen et al., 1993). To form an es organ, the SOP gives rise to two distinct daughter cells, IIa and IIb (see Figure 3A). The IIa cell then divides to produce the hair cell (tricogen) and the socket cell (tormogen), the outer support cells. Shortly after the division of IIa, IIb divides to generate the neuron and the sheath cell (thecogen), the inner cells (Bate, 1978; Hartenstein and Posakony, 1989). A different series of asymmetric divisions allows the SOP of a cho organ (see Figure 3D) to give rise to the neuron and three different support cells (the sheath cell, the cap cell, and the ligament cell) (Bodmer et al., 1989; Brewster and Bodmer, 1995).

Notch and *Delta* have been shown to mediate cell–cell communication in eye development (Cagan and Ready, 1989), muscle development (Corbin et al., 1991), oogenesis (Ruohola et al., 1991; Xu et al., 1992), and neurogenesis (Lehmann et al., 1983) in Drosophila. Homologs of *Notch* and *Delta* have been characterized in various species ranging from worm to human (reviewed by Artavanis-Tsakonas et al., 1995). *Notch* encodes a transmembrane protein with EGF-like repeats in the extracellular domain and tandem ankyrin repeats in the intracellular domain (Wharton et al., 1985; Kidd et al., 1986). The *Notch* product has been postulated to act as a receptor, whereas the *Delta* gene product, another transmembrane protein with EGF repeats (Vaessin et al., 1987; Kopczynski et al., 1988), may function as a ligand for the Notch receptor (Heitzler and Simpson, 1991; Rebay et al., 1991).

Cell–cell interaction is required not only for singling out SOPs, but also for the proper cell fate determination of SOP progeny, as indicated by studies of temperature-sensitive alleles of *Notch* and *Delta*. Reduction of *Notch* or *Delta* function during SOP progeny formation in pupal development causes an adult es organ (bristle) to contain four neurons and no support cells (Hartenstein and Posakony, 1990; Parks and Muskavitch, 1993). A slightly earlier temperature shift, presumably prior to IIa and IIb division, leads to two neurons and two sheath cells in *Delta*^{ts} flies (Parks and Muskavitch, 1993). Thus, it was postulated that *Delta* specifies four distinct progeny cells in a stepwise fashion during the SOP division and the IIb cell division (Parks and Muskavitch, 1993).

Besides cell–cell interaction, a cell-intrinsic factor, the *numb* gene product, is also essential for cell fate determination of SOP progeny. *numb* encodes a protein with a motif called the phosphotyrosine-binding (PTB) domain (Kavanaugh and Williams, 1994) or phosphotyrosine interaction (PID) domain (Bork and Margolis, 1995). Loss of *numb* function transforms the IIb cell into the IIa cell during embryogenesis (Uemura et al., 1989), and overexpression of *numb* results in reciprocal cell fate transformation (Rhyu et al., 1994). Immunocytochemical studies have shown that Numb is asymmetrically localized to one pole of the SOP and segregated to one of the daughter cells (Rhyu et al., 1994; Knoblich et al., 1995). Thus, asymmetrically distributed Numb confers distinct daughter cell fates (Rhyu et al., 1994). Asymmetric divisions of the SOP and its daughter cells that give rise to the adult es organ (bristle) also depend on *numb* function (Rhyu et al., 1994); the cell fate transformation due to overexpression of *numb* via heat shock (*hs-numb*) is similar to the phenotype caused by reduction of *Notch* function. Similarly, asymmetric division of the MP2 neural precursor in the central nervous system (CNS) depends on asymmetric localization of Numb (Spana et al., 1995).

numb exerts its function at least in part via a downstream target gene called *tramtrack* (*ttk*) (Guo et al., 1995). *ttk* encodes two zinc finger proteins due to alternative splicing (Harrison and Travers, 1990; Brown et al., 1991; Read and Manley, 1992). The 69 kDa Ttk protein has been shown to act as a transcriptional repressor in the process of segmentation (Brown et al., 1991; Read et al., 1992). Like *numb*, *ttk* acts as a genetic switch. Loss of *ttk* function causes the SOP daughter cell IIa to be transformed into IIb, a phenotype opposite to the loss of *numb* function phenotype. Overexpression of *ttk*, on the other hand, results in the same cell fate transformation as does loss of *numb* function (Guo et al., 1995). Genetic and immunocytochemical studies indicate that *ttk* is negatively regulated by *numb* (Guo et al., 1995).

Several questions are raised by the dependence of asymmetric division on a gene encoding a cell-intrinsic determinant (Numb) as well as genes such as *Notch* and *Delta*, which mediate cell–cell interaction. How does the information derived from cell-intrinsic signals interface with signals arising from cell–cell interaction? How are these different instructions to the two daughter cells implemented to secure their distinct fates? In this paper, we first show that *Notch* plays a critical role in asymmetric divisions during embryogenesis. We then demonstrate that genetically *Notch* is most likely negatively regulated by *numb*, and biochemically Notch binds to Numb. This direct protein–protein interaction requires either the RAM23 region or the very C-terminal end of Notch and the PTB domain of Numb. Finally, we have identified *ttk* as a downstream target gene of *Notch*. Taken together, we propose that Ttk acts as a readout to integrate the information derived from both cell-intrinsic mechanism and cell–cell communication; the inherited determinant Numb sets up or biases the direction of cell–cell interaction to allow the specification of distinct daughter cell fates during asymmetric divisions.

Results

Reduction of *Notch* Function during SOP Division Causes Support Cells to Be Transformed into Neurons

Although there have been extensive studies of *Notch* function in the process of lateral inhibition to single out a neuronal precursor from a proneural cluster (Campos-Ortega, 1988; Artavanis-Tsakonas and Simpson, 1991; Ghysen et al., 1993), the role of *Notch* in asymmetric divisions of the SOP has only been examined in the adult (Hartenstein and Posakony, 1990), but not in the embryo. Hence, we characterized the embryonic phenotypes due to reduction of *Notch* function or overexpression of *Notch* and showed that they corresponded to reciprocal daughter cell fate transformations.

Elimination of *Notch* function throughout development results in a significant overproduction of neuronal precursors (Lehmann et al., 1983). The general disorganization of the PNS in these null mutants, however, makes it difficult to ascertain any cell fate alterations within the lineage of a single sensory organ. To overcome this difficulty, a temperature-sensitive allele of *Notch* (*Notch*^{ts}) was used. Embryos of *Notch*^{ts} were first raised at the permissive temperature (18°C) to allow the normal development of the embryo and then shifted to the nonpermissive temperature (30°C) at certain developmental stages to disrupt the function of *Notch* (see Experimental Procedures).

SOPs are singled out and then divide asymmetrically during 4–7 hr of embryogenesis at 25°C (Bodmer et al., 1989). Shifting *Notch*^{ts} embryos to the nonpermissive temperature at stages corresponding to 4–5 hr of embryogenesis at 25°C resulted in significant overproduction of neurons in most embryos, as revealed by staining with a neuronal marker MAb22C10 (Zipursky et al., 1984). Using anti-Cut antibody, which recognizes all four cells of the es organ (Blochlinger et al., 1990), we examined the site of emergence of normally a single es organ and found that the number of Cut-positive cells increased by a factor of two to three (data not shown). Thus, most likely supernumerary SOPs emerged from the disruption of *Notch* function at this early stage due to failure of singling out of SOPs.

When *Notch*^{ts} embryos were shifted to the nonpermissive temperature at stages corresponding to 5–7 hr of embryogenesis at 25°C, we observed phenotypes of cell fate transformation within both es and cho lineages. We focused our studies in the dorsal-most region of an abdominal hemisegment, which normally harbors two es organs, and doubly stained the embryos with anti-Cut antibody and MAb22C10. In wild type, there are eight Cut-positive cells and two MAb22C10-positive neurons. Among younger embryos in a *Notch*^{ts} embryo collection, which would have their *Notch* function disrupted at an earlier stage, some had ten to twelve Cut-positive cells; most of these cells expressed MAb22C10. The rest of the embryos had a complement of eight Cut-positive cells, though up to six of these cells also expressed MAb22C10. It thus appears that even after the singling out of the SOPs, the number of neurons within an es organ increases with the disruption of *Notch* function. This could be interpreted as a cell fate transformation during asymmetric divisions of SOP, analogous to what has been observed in the adult bristle formation (Hartenstein and Posakony, 1990).

To test this further, we doubly stained the *Notch*^{ts} embryos with MAb22C10, a neuronal marker, and anti-Prospero antibody, which labels the nuclei of sheath cells in both es and cho organs (Vaessin et al., 1991). We focused our studies on the older embryos in the collection that would have their *Notch* function disrupted at a later stage. In the wild-type embryo, the two

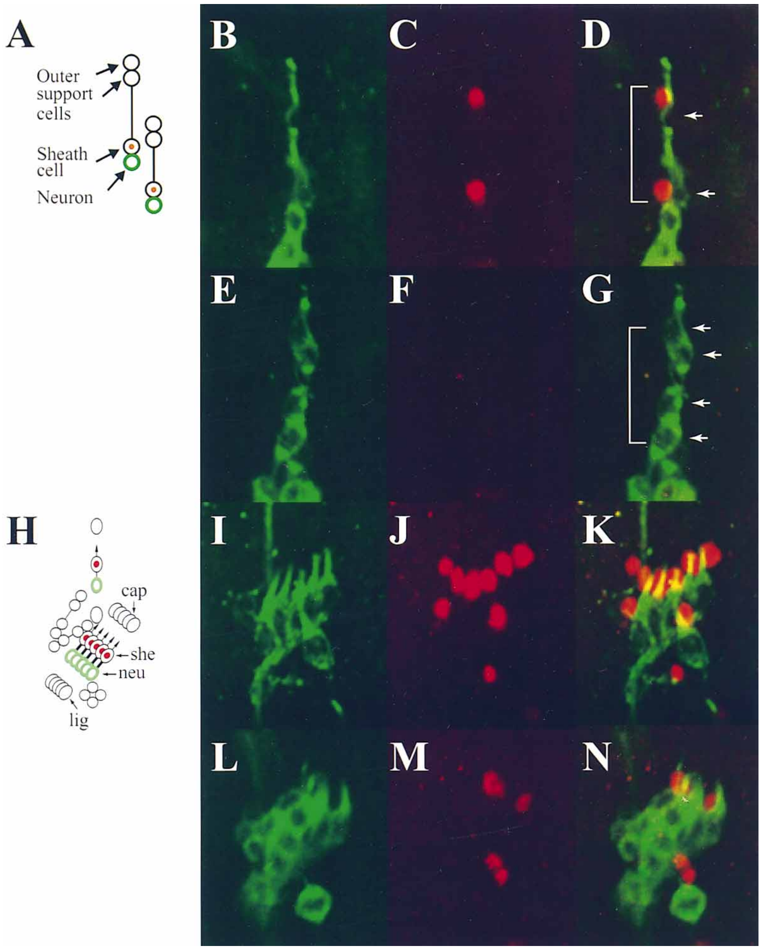

Figure 1. Reduction of *Notch* Function Transforms Sheath Cells to Neurons

(A–D) Two simple es organs at the dorsal-most region of an abdominal segment in wild-type embryo. (A) A schematic drawing of the two isolated es organs, with each circle representing an individual cell. Green circles correspond to the MAb22C10 staining that recognizes the cytoplasm of neurons shown in (B) and (D), whereas red dots correspond to the anti-Prospero staining that labels the nuclei of sheath cells shown in (C) and (D). In wild type, two neurons (B) (two arrows in [D]) and two sheath cells (C) can be observed (bracket in [D]).

(E–G) In *Notch*^{ts}, four neurons ([E] and four arows in [G]) can be detected without the associated sheath cells ([F] and bracket in [G]).

(H) Lateral region of an abdominal segment where five cho organs are aligned side by side. In this region, there are also other cells, one v′ cho organ and three es organs. Similarly to (A), green circles and red dots correspond to the MAb22C10 and anti-Prospero staining respectively. she, sheath cell; neu, neuron; lig, ligament cell; cap, cap cell.

(I–K) In wild type, five cho neurons (I and K) have five sheath cells associated with them (J and K).

(L–N) In *Notch*^{ts}, the number of MAb22C10-positive neurons increases (L and N), while the number of anti-Prospero positive sheath cells decreases (M and N). Moreover, the number of neuronal overproduction is equivalent to that of the sheath cell reduction (N).

isolated dorsal-most es organs have two MAb22C10-positive neurons and two Prospero-expressing sheath cells (Figures 1A–1D). In contrast, the *Notch*^{ts} embryos contained four neurons without associated sheath cells at the corresponding location (Figures 1E–1G), indicating that the sheath cells are transformed into neurons (see Figure 3B). As expected from such a cell fate transformation, staining with anti-Cut antibody revealed four strongly staining cells (hairs or sockets) and four weakly staining cells (neurons or sheath cells) (data not shown). Similar cell fate transformation from MAb22C10-positive neurons to Prospero-positive sheath cells in cho organs was observed (Figures 1L–1N). This cell fate transformation from support cells to neurons in both es and cho organs further substantiates the role of *Notch* in cell fate specification during asymmetric divisions. These embryonic phenotypes were then used to investigate the epistatic genetic relationships between *Notch* and *ttk*, as well as between *Notch* and *numb* (see below).

Overexpression of Activated Notch Results in the Reverse Cell Fate Transformation

When a truncated form of *Notch* (intracellular domain) is ubiquitously overexpressed during the stage when neuronal precursors are singled out, there is a reduction of neuronal precursors (Lieber et al., 1993; Rebay et al., 1993; Struhl et al., 1993; Lyman and Yedvobnick, 1995; Bang et al., 1995). Since these phenotypes are opposite to that due to loss of *Notch* function during the process of singling out of neuronal precursors, it is suggested that the Notch intracellular domain corresponds to a constitutively active form of the Notch receptor (Lieber et al., 1993; Rebay et al., 1993; Struhl et al., 1993).

We used the UAS–GAL4 system (Brand and Perrimon, 1993) and studied the effect of overexpression of the constitutively active form of Notch (Notch^{ACT}) (Doherty et al., 1996) that contains a truncated Notch (with the extracellular domain deleted but with the transmembrane domain and the intracellular domain retained) during SOP divisions. Transgenic flies carrying UAS–Notch^{ACT} were mated to transgenic flies carrying a GAL4 enhancer–trap line, 1407 GAL4, which expresses the reporter gene in all neurons of the PNS (Luo et al., 1994). The embryos carrying UAS–Notch^{ACT} as well as the 1407 GAL4 line would then express Notch^{ACT} primarily in neurons. At the dorsal-most region of abdominal segments of these embryos, anti-Cut staining showed the expression pattern of four strongly stained and four weakly stained cells in approximately 90% of the segments (data not shown). Double labeling of both MAb22C10 and anti-Prospero, however, revealed a cell fate transformation in at least one of these two es organs in about 30%–40% of the abdominal segments examined. A typical example is shown in Figures 2A–2C, in which the es organ on the top contains two anti-Prospero-positive cells but no neurons, suggestive of a transformation

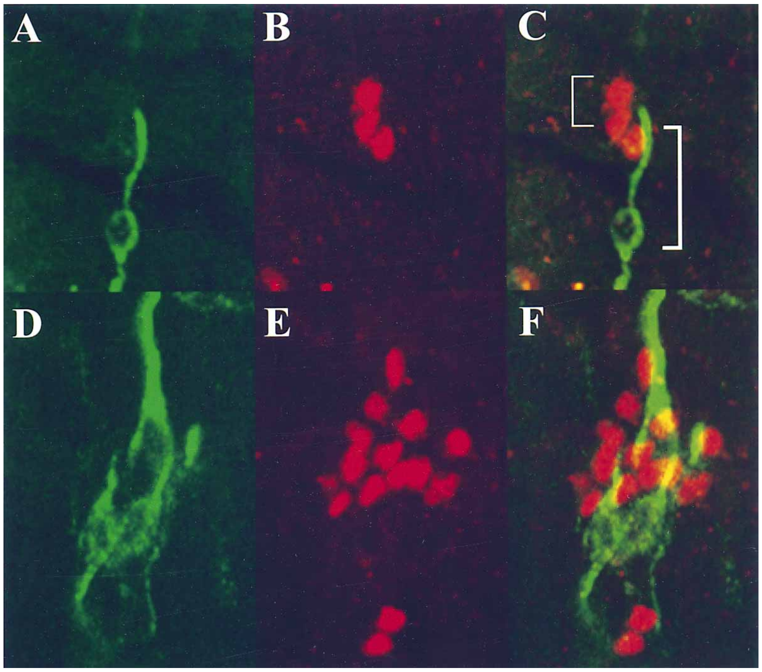

Figure 2. Overexpression of $Notch^{ACT}$ Transforms Neurons to Sheath Cells

(A–C) Of the two dorsal-most es organs, the top es organ (the bracket on the left in [C]) has two anti-Prospero positive sheath cells (B), without the associated MAb22C10-positive neuron (A). (C) The lower es organ (the bracket on the right in [C]) is not transformed. (D–F) In the lateral region of an abdominal segment, the number of anti-Prospero positive sheath cells increases (E and F), while the number of MAb22C10-positive neurons decreases (D and F). Moreover, the overproduction of sheath cells matches in number with the reduction of neurons (F).

from neurons to sheath cells (Figure 3C). The lower es organ, on the other hand, appears normal. In the cho organs at the lateral region, we also observed an increase in the number of Prospero-positive sheath cells and a concurrent decrease in the number of MAb22C10-positive neurons (Figures 2D–2F), indicative of a cell fate transformation from neurons to sheath cells (Figure 3F). In this region, at least three of the five cho organs in most segments showed transformation. Thus, overexpression of $Notch^{ACT}$ results in a cell fate transformation opposite to that caused by reduction of *Notch* function.

Overexpression of the Wild-Type Notch Results in Cell Fate Transformation Similar to That Caused by Activated Notch

If overexpressing the truncated Notch, $Notch^{ACT}$, indeed reveals the activity of the Notch receptor upon activation, it should produce phenotypes qualitatively similar to those due to overexpression of the wild-type Notch ($Notch^{WT}$). It has been reported that overexpression of wild-type Notch via the heat shock promoter does not result in any significant phenotypes (Lieber et al., 1993; Rebay et al., 1993; Fortini et al., 1993). We found that Notch overexpression due to UAS–$Notch^{WT}$ in combination with various GAL4 enhancer–trap lines resulted in a very mild phenotype of transformation of neurons to sheath cells in the embryo (data not shown), but much stronger phenotype in adult es organs (bristles).

Overexpression of $Notch^{ACT}$ in the 109-68 GAL4 enhancer–trap line, which expresses GAL4 in all four cells of an adult es organ, caused 90% of the bristles in the notum to form double sockets (Figure 4A) or, less frequently, triple sockets or four sockets (Figure 4B). The double socket phenotype is similar to the phenotype due to overexpression of the intracellular domain of Notch (Lieber et al., 1993; Rebay et al., 1993; Struhl et al., 1993; Lyman and Yedvobnick, 1995; Bang et al., 1995) and is indicative of a transformation of the hair cell into the socket cell. The four sockets most likely arise from a transformation of the IIb cell into the IIa cell, followed with a transformation of the hair cell into the socket cell (Figure 4E). Overexpression of $Notch^{WT}$ using the same GAL4 enhancer–trap line caused about 20% of the bristles to form double sockets (Figure 4C) or four sockets (Figure 4D). Thus, $Notch^{ACT}$ overexpression causes qualitatively similar, albeit stronger, phenotypes, as compared with those due to overexpression of the wild-type Notch product.

Sensitive Genetic Interaction between *numb* and *Notch*

The effects of reducing *Notch* function on adult bristle formation (Hartenstein and Posakony, 1990) are similar to those due to overexpression of *numb* (Rhyu et al., 1994). To determine whether *Notch* and *numb* function in the same genetic pathway, we first tested to see whether there is a synergistic enhancement of the adult bristle phenotypes due to both reduction of *Notch* function and *numb* overexpression.

Whereas heat shock during larval development causes flies carrying one or two copies of the *hs-numb* transgene to exhibit double hairs, as a result of transformation of the socket cell into the hair cell (Rhyu et al., 1994), without heat shock these flies did not have any bristle phenotype (Figure 5B). Flies carrying only one copy of the functional *Notch* gene, $Notch^{55e11}/+$, also showed no obvious hair or socket abnormalities (Figure 5C). By contrast, even in the absence of any heat shock treatment, $Notch^{55e11}/+;$ *hs-numb*/+ flies had double hairs without sockets at the anterior wing margin (Figure 5D). This phenotype was further enhanced in $Notch^{55e11}/+$ flies that carried two copies of *hs-numb* (Figure 5E). In addition to finding double hairs with no sockets in place of a normal bristle (Figure 5F), we also observed double hairs right next to a normal looking bristle (Figure 5G) and two sets of double hairs right

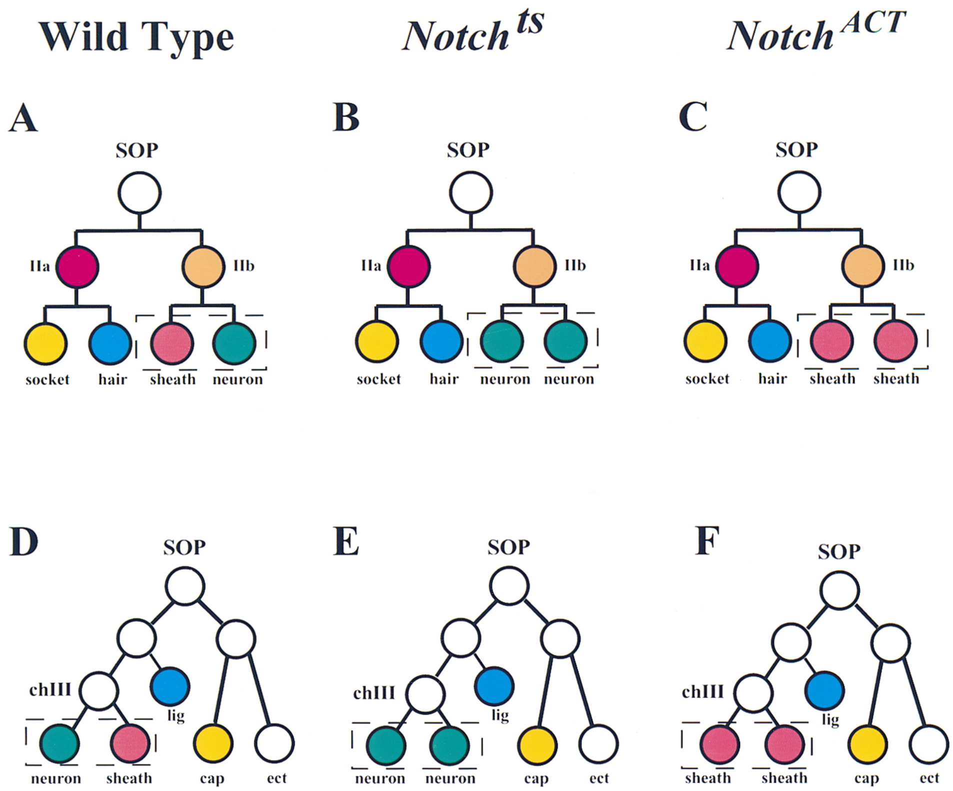

Figure 3. Wild-Type and Proposed SOP Cell Lineages in the Embryo due to Reduction of *Notch* Function or Notch^{ACT} Overexpression

(A–C) Cell lineages for the es organ in wild type (A), *Notch*^{ts} (B), and Notch^{ACT} overexpression (C). Within each lineage, we have boxed two daughters whose fates are indicated to be affected by *Notch*. In the *Notch*^{ts} (B), the sheath cell is transformed into the neuron, yielding two neurons. Following Notch^{ACT} overexpression (C), the neuron is transformed into the sheath cell.

(D–F) Cell lineages for the cho organ in wild type (D), *Notch*^{ts} (E), and Notch^{ACT} overexpression (F). We have modified drawings of the cho lineage according to Brewster and Bodmer (1995). Lig, the ligament cell; ect, the ectodermal cell. In a cho organ, two neurons are produced at the expense of the sheath cell in the *Notch*^{ts} (E). Following Notch^{ACT} overexpression (F), two sheath cells are formed coincident with the loss of the neuron. Both reduction of *Notch* function and overexpression of Notch^{ACT} appear to disrupt the normally asymmetric division, producing two identical daughter cells. Only the effects of reduction of *Notch* function and Notch^{ACT} overexpression on the formation and differentiation of the IIb lineage in the es organ and of the chIII lineage in the cho organ are depicted in this figure; similar cell fate transformations in the IIa lineage and between IIa and IIb have been observed but are not indicated here.

next to each other (Figure 5H). Two independent *Notch* alleles and two independent insertion lines of *hs-numb* were tested, and similar synergistic enhancement of the bristle phenotype was observed (data not shown). Thus, it is unlikely that the observed phenotypes could be due to the site of insertion of the *hs-numb* transgene or unknown genetic background. These bristle phenotypes therefore represent a strong synergistic effect of reducing *Notch* function in the *Notch* heterozygote and slightly elevating Numb expression due to the basal activity of *hs-numb* without heat shock; neither alone generated any detectable phenotype. This suggests, though does not prove, that *numb* and *Notch* function in the same genetic pathway.

Epistatic Relationship between *Notch* and *numb*

To investigate the epistatic relationship between *numb* and *Notch*, we examined whether reduction of *Notch* function is capable of suppressing the phenotypes resulting from loss of *numb* function. Embryos homozygous for the null mutation *numb*^{1} and *Notch*^{ts} were shifted to the nonpermissive temperature at the time of cell fate specification within the lineage of the sensory organs. Compared with the wild type (Figures 6A and 6B), loss of *numb* function reduced the number of neurons to less than 10% of that in the wild type (Uemura et al., 1989), as revealed by the neuronal marker MAb22C10 (Figures 6C and 6D). Reduction of *Notch* function, on the other hand, caused a neuronal overproduction as described earlier (Figures 6E and 6F). The double mutants of *numb* and *Notch*^{ts} showed a range of phenotypes; 10% of the embryos demonstrated a neuronal overproduction approaching the *Notch*^{ts} phenotype (Figures 6G and 6H). The rest of the embryos did not exhibit as extensive an overproduction, but the number of neurons was much larger than that in *numb* null mutants (Figures 6I and 6J). Since a reduction of *Notch* function partially suppressed the *numb* null phenotypes, *Notch* might act downstream of *numb* and appears to be negatively regulated by *numb*. We then examined Numb expression when *Notch* function is disrupted. The asymmetric Numb localization was still present in at least some of the dividing precursor cells in both CNS and PNS of the *Notch*^{55e11} null mutant embryo (data not shown), consistent with *Notch* functioning downstream of *numb*.

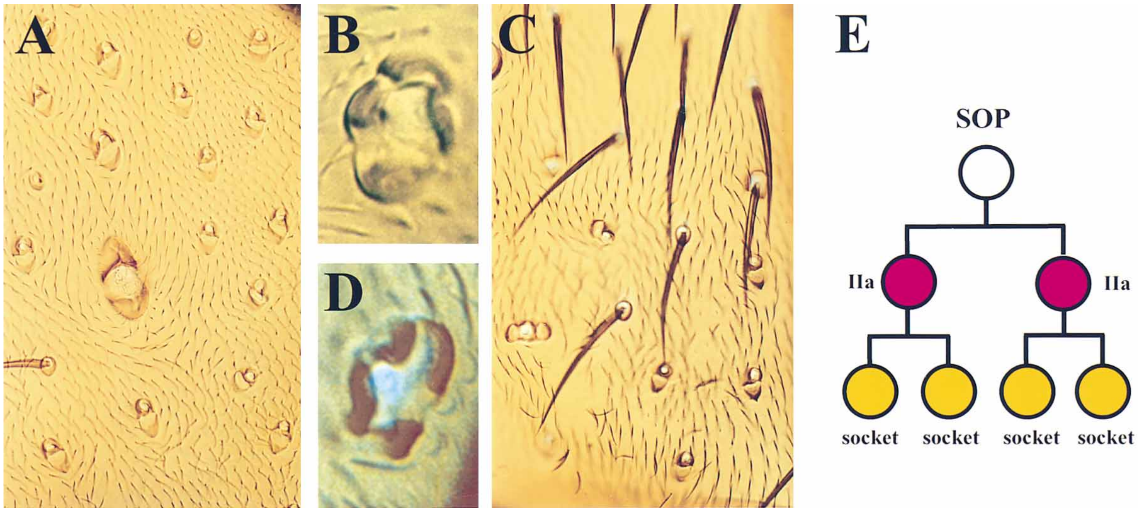

Figure 4. Overexpression of Notch^{ACT} and Notch^{WT} Results in Similar Phenotypes of Multiple Sockets

(A) Double sockets phenotypes of both macrochaetes and microchaetes on the notum due to Notch^{ACT} overexpression.

(B) Four sockets observed in a Notch^{ACT}-expressing fly.

(C) Double sockets due to overexpression of Notch^{WT} observed at a lower frequency.

(D) Four sockets were also seen in Notch^{WT} overexpression.

(E) The proposed lineage for the four socket phenotype observed in both Notch^{ACT} overexpression and Notch^{WT} overexpression. There is a cell fate transformation first from the IIb cell to the IIa cell and then from the hair cell to the socket cell.

Direct Protein–Protein Interaction between Notch and Numb

Since we detected a very strong synergistic interaction in transheterozygous flies carrying *Notch^{55e11}*/+; *hs-numb*/+ (without heat shock) (see Figure 5), we wondered whether there could be direct physical interaction between these two proteins. We assessed this possibility first using the yeast two-hybrid interaction assay (Bartel et al., 1993). Full-length as well as fragments of Numb or Notch were fused with either the LexA DNA-binding domain or the GAL4 transcriptional activation domain. If Numb and Notch physically interact, cotransformation of these two fusion proteins in yeast cells would result in the reconstitution of the "hybrid" transcription factor that can be assessed by the β-galactosidase activity. As described below, this assay revealed interaction between Notch and Numb. These two proteins were then subdivided into fragments to localize the regions involved in this protein–protein interaction.

To identify the portions of the Notch protein involved in the interaction with Numb, we subdivided the Notch intracellular domain into three regions, A, M, and P (Figure 7), and assayed their interactions with Numb individually. Two regions, A and P, interacted with Numb, whereas the M region did not (Figure 7A and Table 1). This interaction appears to be specific, since neither Numb nor Notch-A bound to the control protein Lamin. Moreover, Suppressor of Hairless (Su(H)), a known Notch-binding protein (Tamura et al., 1995; Hsieh et al., 1996), bound to Notch-A but not Notch-P (Table 1). Both regions A and P display a high degree of amino acid conservation among vertebrate Notch homologs (Weinmaster et al., 1991, 1992; Bierkamp and Campos-Ortega, 1993) and include several functional motifs. Region P is at the very C-terminal end of Notch and includes the PEST region; it has been suggested to play an important role in Notch functioning (Xu et al., 1990; Lieber et al., 1993). Region A includes RAM23, a Su(H)-binding region, and ankyrin repeats, a Deltex-binding region (Diederich et al., 1994; Matsuno et al., 1995). Since Su(H) and possibly Deltex are involved in the cell fate decision in the es lineage (Ashburner, 1982; Schweisguth and Posakony, 1994; Matsuno et al., 1995), we wondered whether Numb binds to RAM23 and/or ankyrin repeats. As shown in Table 1 and Figure 7A, Numb bound to RAM23, but not ankyrin repeats of Notch.

Two lines of evidence implicate the N-terminal portion but not the C-terminal portion of Numb in Notch binding. As described later, a glutathione S-transferase (GST) fusion protein of Notch bound to Numb-N but not to Numb-C (Figure 7C). In the yeast two-hybrid assay, Notch-A also showed interaction with Numb-N but not Numb-C (Figure 7B and Table 1). Consistent with the finding, Notch-A also bound to Numb-P, which includes Numb-N and is conserved between Drosophila Numb and mouse Numb (Zhong et al., 1996 [this issue of *Neuron*]; Figure 7B and Table 1). We then tested whether the PTB domain alone, part of Numb-N, is sufficient to confer the binding to Notch-A. Indeed, we detected an interaction as shown in Figure 7B and Table 1.

To confirm the physical interaction between Notch and Numb, we used the in vitro binding assay as an independent test. The Notch intracellular domain was fused to GST, and this fusion protein was expressed in bacteria. The GST–Notch fusion protein was immobilized on glutathione–Sepharose beads and then mixed with in vitro translated ^{35}S-labeled Numb-N or Numb-C. As shown in Figure 7C, Numb-N, but not Numb-C, was retained with GST–Notch on beads. Neither Numb-N nor Numb-C was retained with GST protein alone on the glutathione–Sepharose beads. The ability of Notch intracellular domain to bind Numb-N in vitro as well as in yeast two-hybrid assay suggests that direct protein–

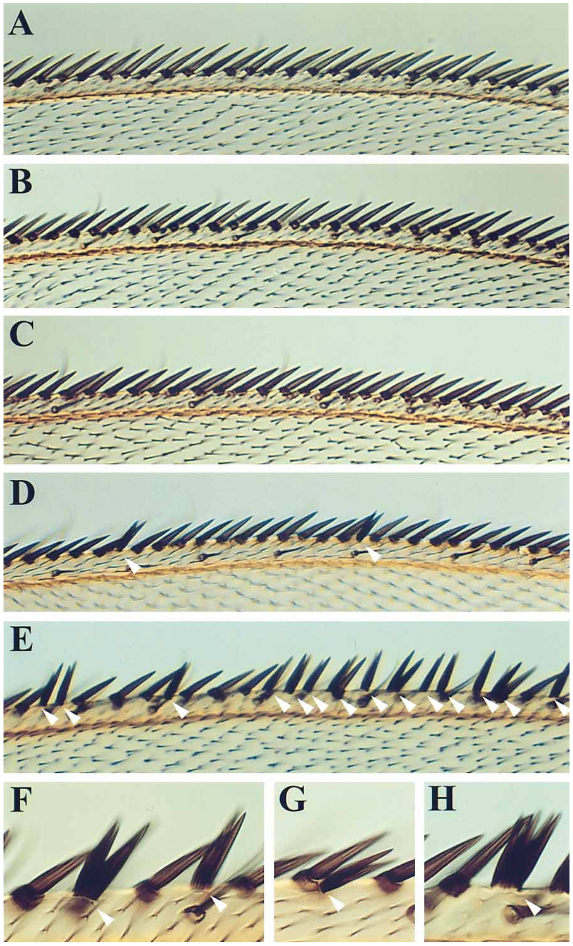

Figure 5. Synergistic Interaction between Reduction of *Notch* Function and Slight Elevation of Numb Expression

The bristles in anterior wing margin of wild type (A), *hs-numb* (B), *Notch*^{55e11}/+ (C), *Notch*^{55e11}/+; *hs-numb*/+ (D), and *Notch*^{55e11}/+; *hs-numb* (E–H). The stout bristles are in focus. Bristles in *hs-numb* (B) and *Notch*^{55e11} /+ (C) are essentially wild type in appearance (A). In *Notch*^{55e11}/+; *hs-numb*/+ (D), there is a phenotype of "double hairs" (arrowhead) at low frequency. In *Notch*^{55e11}/+; *hs-numb*, there is a high frequency of abnormal bristles (E), including "double hairs with no socket" (arrowhead in [F]), "double hairs next to a bristle with a normal looking bristle" (arrowhead in [G]), as well as "four hairs next to one another" (arrowhead in [H]).

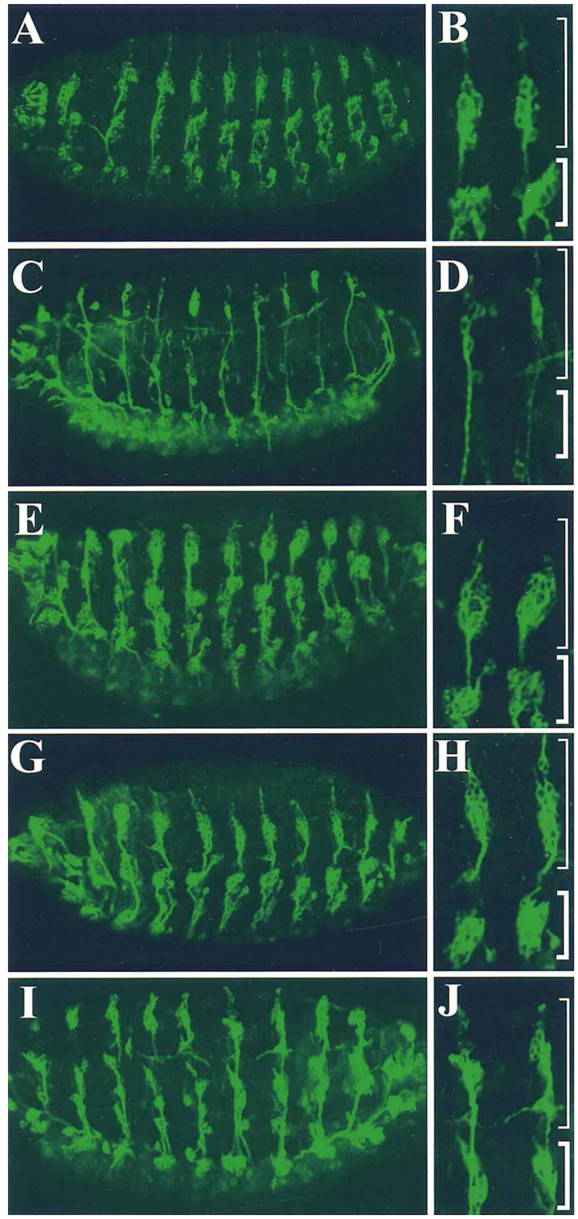

Figure 6. Epistasis of *numb* and *Notch*: Reduction of *Notch* Function Suppresses the Neuronal Phenotype of *numb* Null Allele

MAb22C10 staining of a wild-type embryo (A and B), a *numb*^{1} embryo (C and D), a *Notch*^{ts} embryo (E and F), and the double mutant *numb*^{1}; *Notch*^{ts} embryos (G–J). Whole-mount embryos are in (A), (C), (E), (G), and (I), while the dorsal cluster (thin bracket) and lateral cluster (thick bracket) of two abdominal segments are shown at higher magnification in (B), (D), (F), (H), and (J). Compared with the wild type (A and B), the *numb*^{1} embryo shows a severe reduction in the number of neurons (C and D), whereas the *Notch*^{ts} embryo exhibits a significant increase in the number of neurons (E and F). About 10% of the double mutant *numb*^{1}; *Notch*^{ts} embryos such as the one in (G) and (H) show neuronal overproduction almost to the same extent as *Notch*^{ts} (E and F), whereas the majority of the double mutants such as the one in (I) and (J) shows the "intermediate phenotype," the number of neurons is more than that in *numb*^{1} embryo (C and D), but fewer than that in *Notch*^{ts} embryo (E and F).

protein interaction may provide a molecular mechanism for the negative regulation of *Notch* by *numb*.

ttk Acts Downstream of *Notch*

Since *Notch* and *ttk* are both downstream of *numb* and cause similar phenotypes in the cell fate specification in the es organ lineage, we asked whether *Notch* and *ttk* act in the same genetic pathway and, if so, whether *Notch* acts upstream of *ttk*.

We first examined whether there is any alteration of the Ttk expression due to either reduction of *Notch* function or overexpression of Notch^{ACT}. We focused our investigation on the IIb lineage of the es organ, because Ttk is expressed in the sheath cell that also expresses Prospero, but not in the other daughter cell of IIb, the

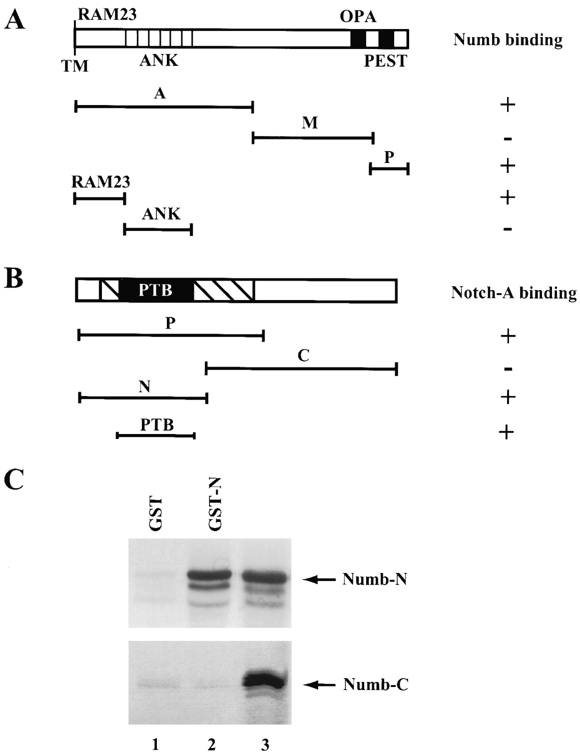

Figure 7. Direct Protein–Protein Interaction between Numb and Notch

(A) and (B) depict the region of Numb and Notch used in two-hybrid assay.

(A) Notch intracellular domain. The RAM23 region, the ankyrin repeats (ANK), the OPA repeats, and the PEST region are indicated. The fusion protein containing regions A, P, or RAM23 binds to Numb (indicated by a plus sign), whereas region M and ANK fail to bind (indicated by a minus sign) (see Table 1).

(B) Numb protein. The conserved region between Drosophila Numb and mouse Numb is illustrated as a striped area and includes the PTB domain. Regions P, N, and PTB, but not C, bind to Notch-A (see Table 1).

(C) In vitro binding of [^{35}S]Numb-N but not [^{35}S]Numb-C to GST–Notch (intracellular domain). [^{35}S]Numb-N or [^{35}S]Numb-C was mixed with the control GST protein (lane 1), or GST–Notch fusion protein (lane 2), as well as glutathione–Sepharose beads. Glutathione–Sepharose beads were used to isolate GST-containing protein complex, which were then analyzed by SDS–PAGE and autoradiography (top, [^{35}S]Numb-N; bottom, [^{35}S]Numb-C). An aliquot of [^{35}S]Numb-N or [^{35}S]Numb-C was run on the same gel (lane 3) to indicate the size of the in vitro translated Numb protein fragments.

neuron that expresses anti-Elav (Guo et al., 1995; Ramaekers et al., submitted). Figures 8A–8F show the double labeling of *Notch*^{ts} embryos using anti-Ttk and anti-Elav antibodies. In the dorsal-most region that normally contains two es organs and hence two neurons, we detected four Elav-expressing neurons (Figure 8B). As shown earlier, the two extra neurons arose from a transformation of the sheath cells into neurons. We found that Ttk was expressed in most of the cells in the epidermis of the *Notch*^{ts} embryo (Figure 8A), but not in neurons, including the supernumerary neurons derived from transformation of sheath cells (Figure 8C). Similar observations were made in the lateral cho organs. None of the neurons, including those sheath cells that had transformed into neurons, were recognized by anti-Ttk antibody (Figures 8D–8F). Thus, *Notch* function is required not only to specify the sheath cell but also to express Ttk in this daughter cell of an asymmetric division.

In a reciprocal experiment, we examined Ttk expression in embryos with Notch^{ACT} overexpression (Figures 8G–8L). In these embryos, we found that often one es organ in the dorsal-most region showed transformation (Figures 8G–8I). Anti-Prospero antibody labeled two sheath cells in this es organ, including one that arose from a transformation of the neuron into a sheath cell (Figure 8H). The superimposition of anti-Ttk staining and anti-Prospero staining reveals that Ttk is expressed in all the sheath cells, including those transformed from neurons (Figure 8I). Similarly, in the cho organs, Ttk is expressed in all the sheath cells, including those that are transformed from neurons (Figures 8J–8L). Thus, activated Notch is able to turn on Ttk expression in cells that normally do not express Ttk. The up- and down-regulation of Ttk expression due to overexpression of activated Notch and reduction of *Notch* function, respectively, indicates that *ttk* is very likely downstream of *Notch* and is positively regulated by *Notch*.

To test this hypothesis further, we asked whether constitutively active Notch requires *ttk* in mediating cell fate transformation. We therefore overexpressed Notch^{ACT} in the *ttk* loss-of-function mutant. Compared with the wild type (see Figure 6A), loss of *ttk* resulted in an overproduction of neurons (Figure 9B), whereas Notch^{ACT} expression resulted in a reduction of neurons (Figure 9A). We found that overexpression of Notch^{ACT} in *ttk* mutant background resulted in an overproduction of neurons (Figure 9C). The extent of this neuronal overproduction is indistinguishable from that due to loss of *ttk* function; without *ttk* function, Notch^{ACT} fails to transform neurons into sheath cells. Therefore, the capability of Notch^{ACT} to transform neurons into sheath cells depends on normal *ttk* function, suggesting that *ttk* acts downstream of *Notch*.

Discussion

Notch Specifies Distinct Daughter Cell Fates during Multiple Asymmetric Divisions in the Formation of Es and Cho Organs

Notch has been shown to play an important role during the process of singling out of SOPs from proneural clusters, a cell fate decision between neuronal fate and epidermal fate. In this paper, by studying the effect of both reduction of Notch function and overexpression of Notch^{ACT} and Notch^{WT}, we show that Notch serves as a binary switch between two daughter cells during SOP divisions in both the es organ and the cho organ.

Notch appears to be important in specifying sheath cell fate in the IIb cell lineage of es organs and the chIII cell lineage of cho organs. Reduction of *Notch* function transforms sheath cells to neurons, whereas overexpression of Notch^{ACT} results in the opposite cell fate transformation in the embryo. Moreover, *Notch* is also involved in the choice between the hair and the socket cell fates. In adult es organs, reduction of *Notch* function in *Notch*^{ts} mutants results in twinning (two hairs without socket) (data not shown), whereas overexpression of either Notch^{ACT} or Notch^{WT} results in double sockets with-

Table 1. Notch and Numb Interact in the Yeast Two-Hybrid Assay

LexA DNA-Binding Domain Fushion	GAL4-Transcriptional Activation Domain Fusion	β-Galactosidase Activity
1. Notch-A	Numb	++
2. Notch-A	Vector alone	−/+
3. Vector alone	Numb	−
4. Lamin	Notch-A	−
5. Lamin	Numb	−
6. Numb	Notch-A	+∼++
7. Numb	Vector alone	−/+
8. Vector alone	Notch-A	−
9. Notch-A	Su(H)	++
10. Vector alone	Su(H)	−
11. Numb	Notch-M	−/+
12. Vector alone	Notch-M	−
13. Notch-P	Numb	+∼++
14. Notch-P	Vector alone	−
15. Notch-P	Su(H)	−
16. Notch-RAM23	Numb	++
17. Notch-RAM23	Vector alone	−/+
18. Notch-RAM23	Su(H)	++
19. Numb	Notch-ANK	−/+
20. Vector alone	Notch-ANK	−/+
21. Numb-P	Notch-A	++
22. Numb-P	Vector alone	−
23. Numb-P	Notch-M	−
24. Notch-A	Numb-P	++
25. Vector alone	Numb-P	−
26. Numb-N	Notch-A	++
27. Numb-N	Vector alone	−
28. Numb-PTB	Notch-A	++
29. Numb-PTB	Vector alone	−
30. Notch-RAM23	Numb-C	−/+
31. Vector alone	Numb-C	−

Plasmids containing fusion protein with LexA–DNA-binding domain and plasmids containing fusion protein with GAL4 activation domain were cotransformed into L40 yeast strain containing the *lacZ* reporter gene under the control of GAL4 elements. The β-galactosidase activity was monitored semiquantitatively by filter lift assay: double plus, turning blue in 1.5 hr; plus, turning blue in 1.5–6 hr; minus/plus, some or all colonies turning blue in 6–24 hr; minus, remaining white over 24 hr.

out hair, the reverse cell fate transformation. Thus, *Notch* seems to promote the socket cell fate in the IIa lineage and the sheath cell fate in the IIb lineage.

Notch also appears to be involved in the choice between the IIa and IIb cell fates during SOP divisions. Reduction of *Notch* function during formation of the adult es organs results in the formation of four neurons at the expense of three support cells (Hartenstein and Posakony, 1990), whereas four sockets are formed in the extreme cases of overexpression of Notch^{ACT} or Notch^{WT}. These phenotypes most likely are produced by first a switch of cell fate between IIa and IIb and then a switch of cell fate between hair and socket or between neuron and sheath. In other words, *Notch* function is required first to specify IIa and then to specify socket (a daughter of IIa) and sheath cell (a daughter of IIb). Reducing or removing *Notch* function results in two IIb cells and their altered divisions leading to the production of four neurons, whereas elevating *Notch* activity leads to two IIa cells, which then divide abnormally and give rise to four sockets. Thus, *Notch* appears to be required for all three asymmetric divisions of the es organ and for at least the final division in the cho organ lineage. The asymmetry is disrupted by either reduction of *Notch* function or *Notch* overexpression, resulting in symmetric divisions.

The Interaction between *Notch* and *numb*

Like *Notch*, *numb* is involved in multiple asymmetric divisions during the formation of a sensory organ (Rhyu et al., 1994). There are several possible scenarios concerning interactions between *numb* and *Notch*. *Notch* could act upstream of *numb*, regulating the asymmetric localization of Numb in the SOP, the segregation of Numb to one of the daughter cells, or Numb activity. Alternatively, Numb, once segregated into one of the daughter cells, could bias subsequent cell–cell interaction by modulating *Notch* signaling. It is also conceivable that *numb* and *Notch* function in parallel to ensure the proper cell fate specification. The possibility that *Notch* acts entirely upstream of *numb* has been ruled out, because a reduction of *Notch* function in double mutants of *Notch*^{ts} and *numb* null mutation partially suppressed the *numb* mutant phenotype. Since these double mutants eliminate *numb* function and yet some of the embryos show overproduction of neurons as do *Notch*^{ts} embryos, the action of *Notch* does not appear to require *numb* gene function. Rather, manifestation of the *numb* mutant phenotype depends on *Notch* gene activity, indicating that *numb* is upstream of *Notch* and negatively regulates *Notch*. In this scenario, loss of *numb* function may lead to an increase of *Notch* activity in the daughter cell that normally inherits Numb. In the double mutant

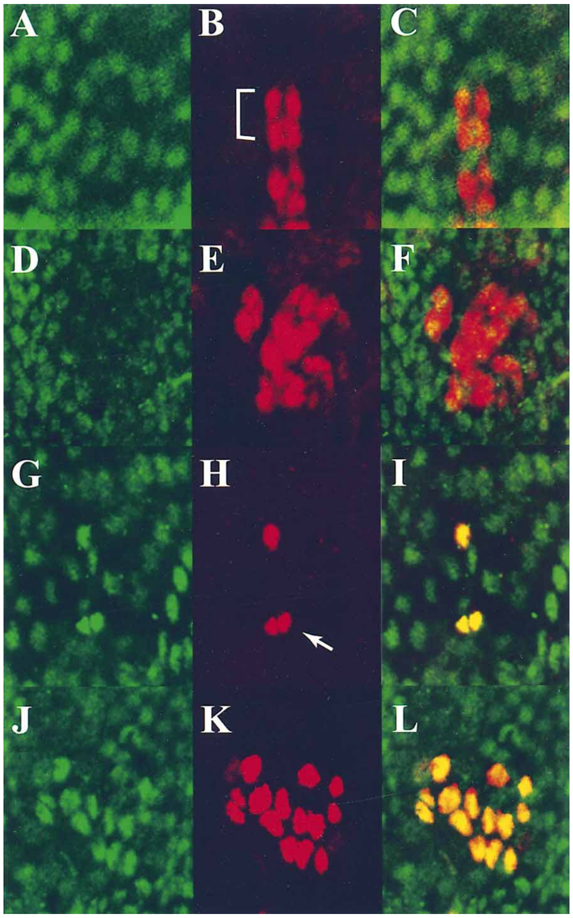

Figure 8. Ttk Expression Is Altered in Opposing Ways in *Notch*^{ts} Mutants and Flies Expressing Notch^{ACT}

(A–C) Double labeling of *Notch*^{ts} embryos using anti-Ttk 88 kDa (A), anti-Elav (B) antibodies, and the overlay of these two images (C) for two dorsal-most es organs. In contrast with the presence of two neurons in wild type, we detected four Elav-expressing neurons (bracketed in [B]). As shown earlier, the two extra neurons arose from a transformation of the sheath cells into neurons. Although Ttk is expressed in most of the cells in the epidermis of the *Notch*^{ts} embryo (A), it is not expressed in any neurons, including the supernumerary neurons derived from transformation of sheath cells (C).

(D–F) The double labeling of *Notch*^{ts} embryos using anti-Ttk 88 kDa (D), anti-Elav (E) antibodies, and the overlay of these two images (F) in five lateral and one v′ cho organs. Similarly, none of the supernumerary neurons (E), including those sheath cells that had transformed into neurons, were recognized by anti-Ttk antibody (D and F).

(G–I) Double labeling of embryos with Notch^{ACT} overexpression using anti-Ttk 88 kDa (G), anti-Prospero (H) antibodies, and the overlay of these two images (I) for the two dorsal-most es organs. Often, one es organ shows transformation. The arrow in (H) points to two sheath cells, including one that is transformed from a neuron. (Top es organ is not transformed.) Ttk is expressed in all sheath cells (G and I), including one that is transformed from a neuron.

(J–L) Double labeling of embryos with Notch^{ACT} overexpression using anti-Ttk 88 kDa (J), anti-Prospero (K) antibodies, and the overlay of these two images (L) for the lateral and v′ cho organs. Similarly, all sheath cells (K), including those that are transformed from neurons, express Ttk (J and L).

that carries the null mutation *numb*^{1} and the hypomorphic allele *Notch*^{ts}, the residual *Notch* activity is likely to be higher than that in the *Notch*^{ts} single mutant. This could account for the milder *Notch* mutant phenotype

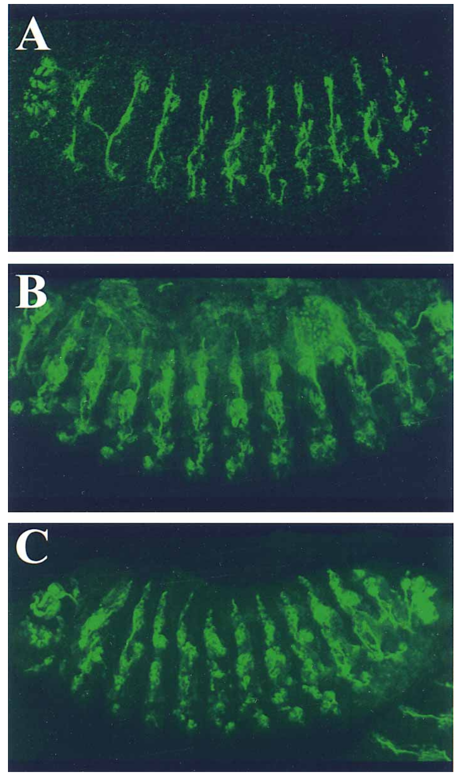

Figure 9. Loss of *ttk* Function Prevents Notch^{ACT} from Transforming Neurons to Sheath Cells

MAb22C10 staining of an embryo expressing Notch^{ACT} (A), a *ttk*^{osn} embryo (B), and a *ttk*^{osn} embryo expressing Notch^{ACT} (C). Compared with the wild type (Figure 6A), there is a reduction in the number of sensory neurons in Notch^{ACT} expressing embryos (A), whereas loss of *ttk* function causes an overproduction of sensory neurons (B). The *ttk*^{osn} embryo that expresses Notch^{ACT} (C) shows the neuronal overproduction phenotype indistinguishable from that due to *ttk* loss of function (B).

in the double mutant. Consistent with the possibility that *numb* is upstream of *Notch*, Numb is still asymmetrically localized in *Notch* null alleles. We also detected a very strong synergistic enhancement of the effects due to a partial loss of *Notch* function and a slight elevation of *numb* expression. While these findings suggest that *Notch* acts downstream of *numb*, as proposed previously by Posakony (1994), Rhyu (1994), and Jan and Jan (1995), they do not exclude the possibility of more complex and elaborate schemes of action of *Notch* and *numb*. For example, *numb* could also be involved in an additional pathway parallel to the *Notch* signaling pathway. Models that incorporate our findings are illustrated in Figure 10. Numb, depicted in red, is asymmetrically distributed to one pole of the SOP during SOP division and segregated into one of the daughter cells so as to bias or set up the direction of cell–cell interaction, causing that cell to adopt the cell fate of IIb. This may be achieved in at least two ways, as indicated by the following two models. In model one (left), cell–cell communication occurs between the two daughter cells. Numb may suppress the Notch (in brown) activity directly, possibly via direct protein–protein interactions between Notch and Numb. Numb may also suppress Notch activity indirectly by regulating Delta (in green)

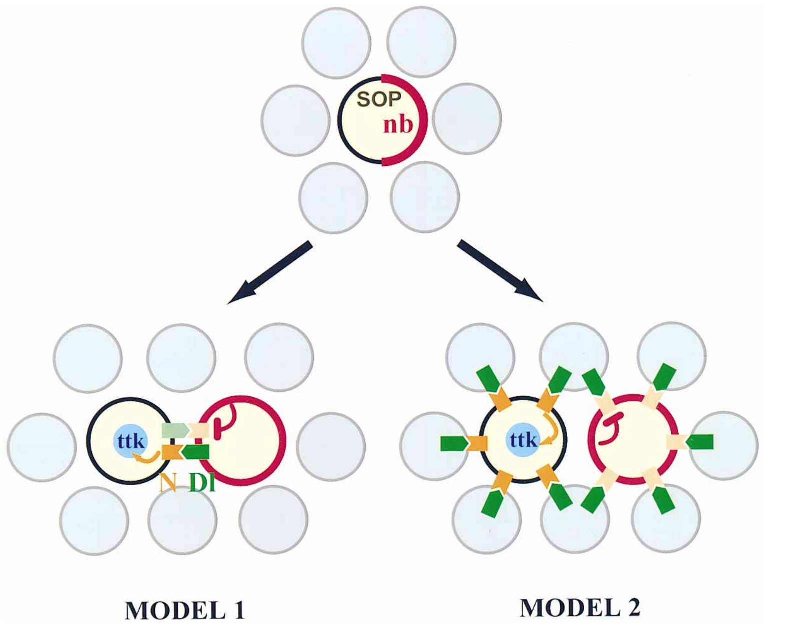

Figure 10. Models for the Interaction among *numb*, *ttk*, and *Notch*

Each circle represents an individual cell. Whereas circles with gray shading represent epidermal cells surrounding the SOP and its daughter cells, circles with yellow shading illustrate the SOP and its daughter cells. The membrane-associated Numb (nb), depicted in red, is segregated into one daughter cell so as to bias or set up the direction of cell–cell interaction. This may be achieved in at least two ways, as indicated by the following two models. In model one (left), cell–cell communication occurs between the two daughter cells. Whereas the Notch activity is illustrated in orange, with the reduced activity shown as the light orange, the Delta activity is demonstrated in green, with the reduced activity in light green. Numb may suppress the Notch activity directly, or indirectly by regulating Delta activity, which renders the Numb-containing daughter cell less effective in receiving signals and more effective in sending signals to its sister cell. In model two (right), the cell–cell signaling mediated by Notch and Delta occurs between SOP daughter cells and the surrounding epidermal cells which express Delta. The suppression of *Notch* by *numb* results in a difference in Notch activity between two daughter cells. These two models are not mutually exclusive. In either model, the Notch activity in the cell without Numb would be sufficiently high to cause Ttk expression (in blue), whereas the Notch activity in the other daughter cell is suppressed by Numb to such an extent that it can no longer induce Ttk expression.

activity. Suppression of Notch activity by Numb renders the Numb-containing daughter cell less effective in receiving signals and more effective in sending signals to its sister cell, so that it adopts the cell fate of IIb and its sister adopts the cell fate of IIa. In model two (right), the cell–cell signaling mediated by Notch and Delta occurs between SOP daughter cells and the surrounding epidermal cells that express Delta. The segregation of Numb into one of the daughter cells causes Notch activity in that cell to be suppressed. The daughter cell that lacks Numb has higher Notch activity and becomes IIa, whereas the daughter cell that has inherited Numb has lower Notch activity, thereby assuming the default cell fate of IIb. These two models are not mutually exclusive. In either model, the Notch activity in the IIa cell would be sufficiently high to cause Ttk expression (in blue), whereas the Notch activity in IIb is suppressed by Numb to such an extent that it can no longer induce Ttk expression.

Our observation of binding of Numb to Notch suggests that Numb may suppress Notch activity through direct protein–protein interaction. One can envision a variety of scenarios as to how this may take place. For instance, the binding of Numb to Notch may prevent Notch from being activated. Alternatively, Numb could inhibit the coupling of activated Notch with its downstream effectors, such as Su(H), or prevent the activation of Su(H) by Notch. Two regions of Notch show interactions with Numb and are highly conserved among Notch homologs in different species. The RAM23 region of Notch physically interacts with Su(H) (Tamura et al., 1995; Hsieh et al., 1996) as well as Numb, whereas Notch-P binds to Numb but not Su(H). Recently, Dishevelled has been shown to interact with Notch C-terminus (Axelrod et al., 1996). It will be interesting to see whether Dishevelled binds to Notch-P, as does Numb.

We have identified the PTB domain of Numb as the Notch-binding domain. PTB domain in other proteins including Shc binds to phosphotyrosine in proteins such as EGF receptor and NGF receptor and is involved in signaling through tyrosine phosphorylation (Kavanaugh and Williams, 1994; Kavanaugh et al., 1995; reviewed by van der Geer and Pawson, 1995; Laminet et al., 1996). This raises the possibility that signaling from Numb to Notch may be regulated by tyrosine phosphorylation. The PTB domain between Drosophila Numb and mouse Numb shares 63.3% amino acid identity (Zhong et al., 1996). We note also that similar protein–protein interaction has been observed between mouse Numb and mouse Notch1 (Zhong et al., 1996), suggesting an evolutionarily conserved mechanism.

ttk Is Regulated by Both *Notch* and *numb*

We have previously demonstrated that *ttk* is involved in multiple asymmetric divisions in both es and cho organs (Guo et al., 1995). Transformation of neurons into sheath cells is caused either by *ttk* overexpression or by Notch^{ACT} overexpression in the presence of normal *ttk* gene function. In the absence of *ttk* function, Notch^{ACT} overexpression fails to transform neurons into sheath cells. This suggests a requirement of *ttk* for the sheath cell fate, though loss of zygotic *ttk* function does not cause sheath cells to be transformed into neurons (Guo et al., 1995), presumably due to the presence of redundant pathways. The dependence of Notch^{ACT} action on *ttk* also indicates that *ttk* is a downstream target of *Notch*. Indeed, whereas Ttk is normally expressed in only one of the IIb daughters, reduction of *Notch* function apparently prevented *ttk* expression in either daughter cell, and overexpression of Notch^{ACT} leads to abnormal *ttk* expression in both daughter cells. Previous studies have shown that *ttk* is negatively regulated by *numb* (Guo et al., 1995). This is consistent with the model shown in Figure 10 in which *ttk* is positively regulated by *Notch*, which is in turn negatively regulated by *numb*. We propose that Ttk, as a transcription factor, acts as a readout to integrate signals derived both from a cell-intrinsic determinant and from cell–cell communication.

Besides *ttk*, other downstream targets of *Notch* have been identified, such as *Su(H)* and *Enhancer of split* (*E(spl)*). Su(H) and Notch exhibit direct protein–protein interaction in the yeast two-hybrid assay (Fortini and Artavanis-Tsakonas, 1994; Tamura et al., 1995; Hsieh et al., 1996). In tissue culture cells, the binding of Delta and Notch allows the translocation of Su(H) from the cytoplasm into the nucleus (Fortini and Artavanis-Tsakonas, 1994). Su(H) then acts as a transcriptional activator to regulate directly *E(spl)* (Lecourtois and Schweiguth, 1995; Bailey and Posakony, 1995), which is a gene complex of seven related genes that encode basic helix-loop-helix proteins characteristic of transcriptional regulators (Delidakis and Artavanis-Tsakonas, 1992; Knust et al., 1992). It is interesting to note that the overexpression of $Notch^{ACT}$ or *Su(H)* results in phenotypes that are slightly different from the phenotype due to the overexpression of *E(spl)*. Whereas overexpression of $Notch^{ACT}$ and *Su(H)* results in predominantly double sockets (Schweisguth and Posakony, 1994), overexpression of the m5 and m8 genes in the *E(spl)* complex results in duplicated bristles, double sockets as well as other aberrant outer support cells (Tata and Hartley, 1995). The phenotype due to overexpression of *ttk* is similar to that of overexpression of *E(spl)* (Guo et al., 1995; Ramaekers et al., submitted). It would be interesting to examine the interaction between *ttk*, *Su(H)*, and *E(spl)* and to investigate how *Notch* signaling eventually leads to the cell fate decision.

Experimental Procedures

Genetics and Drosophila Strains

Drosophila strains were raised on standard cornmeal-yeast agar medium at room temperature or 25°C except those with $Notch^{ts}$ (18°C).

For overexpressing the $Notch^{ACT}$ in *ttk* mutant background, males with a genotype of *yw; 1407 GAL4; ttk*^{osn}*/TM6, Ubx* were mated with females with a genotype of *yw; UAS–Notch*^{ACT} *ttk*^{osn}*/TM3, Sb*. The embryos were collected and stained with markers. All the *ttk* mutant should also have $Notch^{ACT}$ expression driven by *1407 GAL4*. The *ttk* mutant embryos can be unambiguously identified as those embryos that have defects in head involution and dorsal closure (Guo et al., 1995).

For examining the dosage-sensitive interaction between *Notch*^{55e11}/+ and *hs-numb*, *w*^{a}*Notch*^{55e11}*/FM6* females were crossed to *w* males or *yw, hs-numb* (line 2.4C) males, respectively. Female progeny with *Bar*$^{+}$ eye shape, thus with a genotype of *w*^{a}*Notch*^{55e11}/+ or *w*^{a}*Notch*^{55e11}/+; *hs-numb*/+, were collected, and their phenotypes were examined. For examining the phenotype of the *w*^{a}*Notch*^{55e11}/+; *hs-numb* flies, *w*^{a}*Notch*^{55e11}/+; *hs-numb/CyO* stock was first constructed. *w*^{a}*Notch*^{55e11}/+; *hs-numb* flies were unambiguously identified as those with wing notching (due to the haploinsufficiency of the *Notch*^{55e11} allele) and straight wings. All crosses described above were set up in parallel. The same experiment was performed using homozygotes of *nd*^{Grell2}, an independent viable *Notch* allele, and one or two copies of *hs-numb* (line 11.1) located on the third chromosome. Similar synergistic enhancement was observed (see text).

For constructing the double mutants between *numb* and *Notch*, we constructed flies with a genotype of *wNotch*^{ts}*, numb*^{1} *pr cn Bc/CyO*, P[*T8–lacZ*]. Embryos from this stock were collected and double labeled with MAb22C10 and anti-Numb or anti-β-galactosidase antiserum. Double mutants were identified as those embryos that show no anti-Numb epidermal staining nor anti-β-galactosidase staining.

For examining the numb localization, 4.5–6.5 hr embryos of *Notch*^{55e11}*/FM6* flies were doubly stained using both anti-Asense and anti-Numb antibodies. *Notch* mutant embryos were identified as those embryos that have overproduction of neurons in the CNS.

Temperature Shift of *Notch*^{ts} Embryos

wNotch^{ts} stock was kept at 18°C. The developmental time in 18°C was estimated to be twice the amount of time at 25°C (Giniger et al., 1993). Embryos collected between 8–14 hr at 18°C were put into either a 30°C incubator, or a water bath at 32°C. Embryos were kept at these nonpermissive temperatures for 5 hr. After the temperature shift, embryos were placed in an 18°C incubator for 0.5–4 hr prior to fixation.

Immunofluorescence and Confocal Microscopy

Embryos were fixed and stained as described by Guo et al. (1995). A Zeiss microscope and a Bio-Rad MRC-600 confocal were used to view and obtain the images. All the antibodies used in this study have been described previously: MAb22C10 (Zipursky et al., 1984), rat or rabbit anti-Cut antibody (Blochlinger et al., 1990), anti-Prospero antibody (Vaessin et al., 1991), anti-Asense antibody (Brand et al., 1993), guinea pig anti-Ttk 88 kDa antibodies (Read et al., 1992), anti-Elav (Mab44C11) (Bier et al., 1988), and anti-Numb antibody (Rhyu et al., 1994).

Adult nota and wings were dissected from flies in 80% isopropanol and mounted in Hoyer's solution (Ashburner, 1989).

Yeast Two-Hybrid Interaction Assay

Fragments of the *Notch* cDNA were inserted into either the pBHA vector (obtained from C. T. Chien), pGAD424 vector, or pGADGH vector (obtained from Clontech). Region A includes the area from the SalI site to the PstI site (amino acids 1792–2269), or the area from the beginning of the intracellular domain to the EcoRI site (amino acids 1766–2263). Region M spans the area from the EcoRI site (amino acid 1767) to the SalI site (amino acid 2612). Region P includes the C-terminal fragment from amino acid 2602 to the end of the protein. RAM23 construct contains amino acids 1766–1897, whereas the ANK construct amino acids 1895–2109. The amino acid coordinates shown here are as in Wharton et al. (1985). The *numb* full-length cDNA was inserted into either the pGAD vector or the pBHA vector (generated by C. T. Chien). Numb-P starts from the beginning of the protein to the PstI site (amino acids 1–330). Numb-N spans amino acids 1–223 (the BamHI site), whereas Numb-C spans amino acids 224 to the end of the protein. pBHA Numb-N, pBHA Numb-C, and pBHA Numb-P were obtained from C. T. Chien and S. W. Wang. Full-length Su(H) was cloned into pGAD424 (generated by S. W. Wang).

The yeast transformation was performed according to Bartel et al. (1993). The β-galactosidase enzymatic activity was evaluated by filter-lifting assay. Nitrocellulose filters were used to lift up the colonies, and then frozen in liquid nitrogen. Subsequently, the filters were overlaid on the Whatman paper presoaked with 5–45 μl of X-Gal in 3 ml of Z buffer. Colonies producing β-galactosidase turned blue with time. For each set of experiments, a single colony from the transformants carrying either the two fusion proteins to be tested or proper controls (Figure 7B) was then streaked to the same plate and assayed again to ensure the equivalence of experimental conditions.

GST Fusion Proteins and In Vitro Protein Binding Assays

The whole region of Notch intracellular domain (1766–2703) was fused in frame with GST in GST-4T vector (Pharmacia). For protein expression, overnight bacterial culture was diluted and grown until OD at 260 nm reached 1.0. After adding IPTG, the culture was grown for another 3–7 hr. The bacteria were then harvested and lysed by sonication in PBS in the presence of protease inhibitors (aprotinin, leupeptin and pepstatin). Commassie blue staining showed the production of the protein whose size corresponds to the predicted fusion between GST and Notch intracellular domain.

^{35}S-labeled Numb was synthesized in vitro from pNAC-Numb-N or pNAC-Numb-C (generated by C. P. Shen from the same regions of Numb used in two-hybrid assays) using the TNT-coupled rabbit reticulocyte lysate system (Promega); 10 μl of this lysate containing ^{35}S-labeled Numb-N or Numb-C was mixed with GST fusion proteins immobilized on the glutathione–Sepharose 4B beads (Pharmacia), and the mixture was incubated at room temperature for 1–2 hr. The beads were then washed with cold 0.05% Triton X-100 in PBS and collected by centrifugation. To release the bound protein, the beads were boiled in gel sample buffer and analyzed by SDS–polyacrylamide gel electrophoresis (SDS–PAGE) followed by autoradiography.

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References

Artavanis-Tsakonas, S., and Simpson, P. (1991). Choosing a cell fate: a view from the Notch locus. Trends Genet. *7*, 403–408.

Artavanis-Tsakonas, S., Matsuno, K., and Fortini, M.E. (1995). Notch signaling. Science *268*, 225–232.

Ashburner, M. (1982). The genetics of a small autosomal region of *Drosophila melanogaster* containing the structural gene for alcohol dehydrogenase III: hypomorphic mutations affecting the expression of *Hairless*. Genetics *101*, 447–459.

Ashburner, M. (1989). *Drosophila*: A Laboratory Handbook (Cold Spring Harbor, New York: Cold Spring Harbor Laboratory Press).

Axelrod, J.D., Matsuno, K., Artavanis-Tsakonas, S., and Perrimon, N. (1996). Interaction between wingless and Notch signaling pathways mediated by Dishevelled. Science *271*, 1826–1832.

Bailey, A., and Posakony, J. (1995). Suppressor of Hairless directly activates transcription of *Enhancer of split* complex genes in response to Notch receptor activity. Genes Dev. *9*, 2609–2622.

Bang, A., Bailey, A., and Posakony, J. (1995). Hairless promotes stable commitment to the sensory organ precursor cell fate by negatively regulates the activity of the *Notch* signaling pathway. Dev. Biol. *172*, 479–494.

Bartel, P., Chien, C.T., Sternglanz, R., and Fields, S. (1993). Using the two-hybrid system to detect protein–protein interactions. In Cellular Interactions in Development: A Practical Approach, D.A. Hartley, ed. (Oxford: Oxford University Press), pp. 153–179.

Bate, C.M. (1978). Development of sensory systems in arthropods. In Handbook of Sensory Physiology, M. Jacobson, ed. (New York: Springer-Verlag), pp. 1–53.

Bier, E., Ackerman, L., Barbel, S., Jan, L.Y., and Jan, Y.N. (1988). Identification and characterization of a neuron-specific nuclear antigen in *Drosophila*. Science *240*, 913–916.

Bierkamp, C., and Campos-Ortega, J.A. (1993). A zebrafish homologue of the *Drosophila* neurogenic gene Notch and its pattern of transcription during early embryogenesis. Mech. Dev. *43*, 87–100.

Blochlinger, K., Bodmer, R., Jan, L.Y., and Jan, Y.N. (1990). Patterns of expression of Cut, a protein required for external sensory organ development in wild-type and *cut* mutant *Drosophila* embryos. Genes Dev. *4*, 1322–1331.

Bodmer, R., and Jan, Y.N. (1987). Morphological differentiation of the embryonic peripheral neurons in *Drosophila*. Roux's Arch. Dev. Biol. *196*, 69–77.

Bodmer, R., Carretto, R., and Jan, Y.N. (1989). Neurogenesis of the peripheral nervous system in Drosophila melanogaster embryos: DNA replication patterns and cell lineages. Neuron *3*, 21–32.

Bork, P., and Margolis, B. (1995). A phosphotyrosine interaction domain. Cell *80*, 693–694.

Brand, A.H., and Perrimon, N. (1993). Targeted gene expression as a means of altering cell fates and generating dominant phenotypes. Development *118*, 401–415.

Brand, M., Jarman, A.P., Jan, L.Y., and Jan, Y.N. (1993). *asense* is a *Drosophila* neural precursor gene and is capable of initiating sense organ formation. Development *119*, 1–17.

Brewster, R., and Bodmer, R. (1995). Origin and specification of type II sensory neurons in *Drosophila*. Development *121*, 2923–2936.

Brown, J.L., Sonoda, S., Ueda, H., Scott, M., and Wu, C. (1991). Repression of the *Drosophila fushi tarazu* (*ftz*) segmentation gene. EMBO J. *10*, 665–674.

Cagan, R.L., and Ready, D.F. (1989). *Notch* is required for successive cell decisions in the developing *Drosophila* retina. Genes Dev. *3*, 1099–1112.

Campos-Ortega, J.A. (1988). Cellular interactions during early neurogenesis in *Drosophila melanogaster*. Trends Neurosci. *11*, 400–405.

Campos-Ortega, J.A., and Hartenstein, V. (1985). The Embryonic Development of *Drosophila melanogaster* (New York: Springer-Verlag).

Campuzano, S., and Modolell, J. (1992). Patterning of the *Drosophila* nervous system: the *achaete–scute* gene complex. Trends Genet. *8*, 202–208.

Corbin, V., Michelson, A.M., Abmayr, S.M., Neel, V., Alcamo, E., Maniatis, T., and Young, M.W. (1991). A role for the Drosophila neurogenic genes in mesoderm differentiation. Cell *67*, 311–323.

Delidakis, C., and Artavanis-Tsakonas, S. (1992). The *Enhancer of split* [*E(spl)*] locus of *Drosophila* encodes seven independent helix-loop-helix proteins. Proc. Natl. Acad. Sci. USA *89*, 8731–8735.

Diederich, R.J., Matsuno, K., Hing, H., and Artavanis-Tsakonas, S. (1994). Cytosolic interaction between deltex and Notch ankyrin repeats implicates deltex in the Notch signaling pathway. Development *120*, 473–481.

Doherty, D., Feger, G., Younger-Shepherd, S., Jan, L.Y., and Jan, Y.N. (1996). Delta is a ventral to dorsal signal complementary to Serrate, another Notch ligand, in *Drosophila* wing formation. Genes Dev. *10*, 421–434.

Fortini, M.E., and Artavanis-Tsakonas, S. (1994). The Suppressor of Hairless protein participates in Notch receptor signaling. Cell *79*, 273–282.

Fortini, M.E., Rebay, I., Caron, L.A., and Artavanis-Tsakonas, S. (1993). An activated *Notch* receptor blocks cell fate commitment in the developing *Drosophila* eye. Nature *365*, 555–557.

Ghysen, A., and Dambly-Chaudiere, C. (1988). From DNA to form: the *achaete–scute* complex. Genes Dev. *7*, 723–733.

Ghysen, A., Dambly-Chaudiere, C., Aceves, E., Jan, L.Y., and Jan, Y.N. (1986). Sensory neurons and peripheral pathways in *Drosophila* embryos. Roux's Arch. Dev. Biol. *195*, 281–289.

Ghysen, A., Dambly-Chaudiere, C., Jan, L.Y., and Jan, Y.N. (1993). Cell interactions and gene interactions in peripheral neurogenesis. Genes Dev. *7*, 723–733.

Giniger, E., Jan, L.Y., and Jan, Y.N. (1993). Specifying the path of the intersegmental nerve of the *Drosophila* embryo: a role for *Delta* and *Notch*. Development *117*, 431–440.

Guo, M., Bier, E., Jan, L.Y., and Jan, Y.N. (1995). *tramtrack* acts downstream of *numb* to specify distinct daughter cell fates during asymmetric cell divisions in the Drosophila PNS. Neuron *14*, 913–925.

Harrison, S., and Travers, A. (1990). The *tramtrack* gene encodes a *Drosophila* finger protein that interacts with the *ftz* transcriptional regulatory region and shows a novel embryonic expression pattern. EMBO J. *9*, 207–216.

Hartenstein, V., and Posakony, J.W. (1989). Development of adult sensilla on the wing and notum of *Drosophila melanogaster*. Development *107*, 389–405.

Hartenstein, V., and Posakony, J.W. (1990). A dual function of the *Notch* gene in *Drosophila* sensillum development. Dev. Biol. *142*, 13–30.

Heitzler, P., and Simpson, P. (1991). The choice of cell fate in the epidermis of Drosophila. Cell *64*, 1083–1092.

Horvitz, H.R., and Herskowitz, I. (1992). Mechanisms of asymmetric cell division: two Bs or not two Bs, that is the question. Cell *68*, 237–255.

Hsieh, J.J.-D., Henkel, T., Salmon, P., Robey, E., Peterson, M.G., and Hayward, S.D. (1996). Truncated mammalian Notch1 activates CBF1/RBPJk-repressed genes by a mechanism resembling that of Epstein–Barr virus EBNA2. Mol. Cell. Biol. *16*, 952–959.

Jan, Y.N., and Jan L. Y. (1995). Maggot's hair and bug's eye: role of cell interactions and intrinsic factors in cell fate specification. Neuron *14*, 1–5.

Kavanaugh, W.M., and Williams, L.T. (1994). An alternative to SH2 domains for binding tyrosine-phosphorylated proteins. Science *266*, 1862–1865.

Kavanaugh, W.M., Turck, C.W., and Williams, L.T. (1995). PTB domain binding to signaling proteins through a sequence motif containing phosphotyrosine. Science *268* , 1177–1179.

Kidd, S., Kelley, M.R., and Young, M.W. (1986). Sequence of the *Notch* locus of *Drosophila melanogaster*: relationship of the encoded protein to mammalian clotting and growth factors. Mol. Cell. Biol. *6*, 3094–3108.

Knoblich, J., Jan, L.Y., and Jan, Y.N. (1995). Localization of Numb and Prospero via a novel mechanism for asymmetric protein segregation. Nature *377*, 624–627.

Knust, E., Schrons, H., Grawe, F., and Campos-Ortega, J.A. (1992). Seven genes of the *Enhancer of split* complex of *Drosophila melanogaster* encode helix-loop-helix proteins. Genetics *132*, 505–518.

Kopczynski, C.C., Alton, A.K., Fechtel, K., Kooh, P.J., and Muskavitch, M.A.T. (1988). *Delta*, a *Drosophila* neurogenic gene, is transcriptionally complex and encodes a protein related to blood coagulation factors and epidermal growth factor of vertebrates. Genes Dev. *2*, 1723–1735.

Laminet, A.A., Apell, G., Conroy, L., and Kavanaugh, W.M. (1996). Affinity, specificity, and kinetics of the interaction of the SHC phosphotyrosine binding domain with asparagine-X-X-phosphotyrosine motifs of growth factor receptors. J. Biol. Chem. *271*, 264–269.

Lecourtois, M., and Schweiguth, F. (1995). The neurogenic Suppressor of Hairless DNA-binding protein mediates the transcriptional activation of the *Enhancer of split* complex genes triggered by Notch signaling. Genes Dev. *9*, 2598–2608.

Lehmann, R., Jimenez, F., Dietrich, U., and Campos-Ortega, J.A. (1983). On the phenotype and development of mutants of early neurogenesis in *Drosophila melanogaster*. Roux's Arch. Dev. Biol. *192*, 62–74.

Lieber, T., Kidd, S., Alcamo, E., Corbin, V., and Young, M.W. (1993). Antineurogenic phenotypes induced by truncated Notch proteins indicate a role in signal transduction and may point to a novel function for Notch in nuclei. Genes Dev. *7*, 1949–1965.

Luo, L., Liao, Y.J., Jan, L.Y., and Jan, Y.N. (1994). Distinct morphogenetic functions of similar small GTPases: *Drosophila Drac1* is involved in axonal outgrowth and myoblast fusion. Genes Dev. *8*, 1787–1802.

Lyman, D. and Yedvobnick, B. (1995). Drosophila *Notch* receptor activity suppresses Hairless function during adult external sensory organ development. Genetics *141*, 1491–1505.

Matsuno, K., Diederich, R.J., Go, M.J., Blaumueller, C.M., and Artavanis-Tsakonas, S. (1995). Deltex acts as a positive regulator of Notch signaling through interactions with the Notch ankyrin repeats. Development *121*, 2633–2644.

Parks, A., and Muskavitch, M.A.T. (1993). *Delta* function is required for bristle organ determination and morphogenesis in *Drosophila*. Dev. Biol. *157*, 484–496.

Posakony, J.W. (1994). Nature versus nurture: asymmetric cell divisions in Drosophila bristle development. Cell *76*, 415–418.

Read, D., and Manley, J. (1992). Alternatively spliced transcripts of the *Drosophila tramtrack* gene encode zinc finger proteins with distinct DNA binding specificities. EMBO J. *11*, 1035–1044.

Read, D., Levine, M., and Manley, J. (1992). Ectopic expression of the *Drosophila tramtrack* gene results in multiple embryonic defects, including repression of *even-skipped* and *fushi tarazu*. Mech. Dev. *38*, 183–196.

Rebay, I., Fleming, R.J., Fehon, R.G., Cherbas, L., Cherbas, P., and Artavanis-Tsakonas, S. (1991). Specific EGF repeats of Notch mediate interactions with Delta and Serrate: implications for Notch as a multifunctional receptor. Cell *67*, 687–699.

Rebay, I., Fehon, R.G., and Artavanis-Tsakonas, S. (1993). Specific truncations of Drosophila Notch define dominant activated and dominant negative forms of the receptor. Cell *74*, 319–329.

Rhyu, M. (1994). How sister cells acquire different fates: the role of *numb* in asymmetric divisions. PhD thesis, University of California, San Francisco, California.

Rhyu, M., Jan, L., and Jan, Y.N. (1994). Asymmetric distribution of numb protein during division of the sensory organ precursor confers distinct fates to daughter cells. Cell *76*, 477–491.

Ruohola, H., Bremer, K.A., Baker, D., Swedlow, J.R., Jan, L.Y., and Jan, Y.N. (1991). Role of neurogenic genes in establishment of follicle cell fate and oocyte polarity during oogenesis in Drosophila. Cell *66*, 433–449.

Schweisguth, F., and Posakony, J.W. (1994). Antagonistic activities of *Suppressor-of-Hairless* and *Hairless* control alternative cell fates in the *Drosophila* adult epidermis. Development *120*, 1433–1441.

Spana, E., Kopczynski, C., Goodman, C., and Doe, C. (1995). Asymmetric localization of numb autonomously determines sibling neuron identity in the *Drosophila* CNS. Development *121*, 3187–3195.

Struhl, G., Fitzgerald, K., and Greenwald, I. (1993). Intrinsic activity of the *lin-12* and *Notch* intracellular domains in vivo. Cell *74*, 331–345.

Tamura, K., Taniguchi, Y., Minoguchi, S., Saka, T., Tun, T., Furukawa, T., and Honjo, T. (1995). Physical interaction between a novel domain of the receptor Notch and the transcription factor RBP-Jk/ Su(H). Curr. Biol. *5*, 1416–1423.

Tata, F., and Hartley, D.A. (1995). Inhibition of cell fate in Drosophila by *Enhancer of split* genes. Mech. Dev. *51*, 305–315.

Uemura, T., Shepherd, S., Ackerman, L., Jan, L.Y., and Jan, Y.N. (1989). *numb*, a gene required in determination of cell fate during sensory organ formation in Drosophila embryos. Cell *58*, 349–360.

Wharton, K.A., Johansen, K.M., Xu, T., and Artavanis-Tsakonas, S. (1985). Nucleotide sequence from the neurogenic locus *Notch* implies a gene product that shares homology with proteins containing EGF-like repeats. Cell *43*, 567–581.

Vaessin, H., Bremer, K., Knust, E., and Campos-Ortega, J.A. (1987). The neurogenic gene *Delta* of *Drosophila melanogaster* is expressed in neurogenic territories and encodes a putative transmembrane protein with EGF-like repeats. EMBO J. *6*, 3431–3440.

Vaessin, H., Grell, E., Wolff, E., Bier, E., Jan, L.Y., and Jan., Y.N. (1991). *prospero* is expressed in neuronal precursors and encodes a nuclear protein that is involved in the control of axonal outgrowth in Drosophila. Cell *67*, 941–953.

van der Geer, P., and Pawson, T. (1995). The PTB domain: a new protein module implicated in signal transduction. Trends Biochem. Sci. *20*, 277–280.

Weinmaster, G., Roberts, V.J., and Lemke, G. (1991). A homolog of *Drosophila Notch* expressed during mammalian development. Development *113*, 199–205.

Weinmaster, G., Roberts, V.J., and Lemke, G. (1992). Notch2: a second mammalian Notch gene. Development *116*, 931–941.

Xu, T., Rebay, I., Fleming, R.J., Scottgale, T.N., and Artavanis-Tsakonas, S. (1990). The *Notch* locus and the genetic circuitry involved in early *Drosophila* neurogenesis. Genes Dev. *4*, 464–475.

Xu, T., Caron, L.A., Fehon, R.G., and Artavanis-Tsakonas, S. (1992).

The involvement of the *Notch* locus in *Drosophila* oogenesis. Development *115*, 913–922.

Zhong, W.M., Feder, J.N., Jiang, M.M., Jan, L.Y., and Jan, Y.N. (1996). Asymmetric localization of a mammalian Numb homolog during mouse cortical neurogenesis. Neuron *17*, this issue.

Zipursky, S.L., Venkatesh, T.R., Teplow, D.B., and Benzer, S. (1984). Neuronal development in the *Drosophila* retina: monoclonal antibodies as molecular probes. Cell *36*, 15–26.